Strategic EV Market Expansion & Growth Analysis – India

Strategic EV Market Expansion & Growth Analysis – India | Automotive Industry

Let's Begin

By Ishwari Attarde

Strategic EV Market Expansion & Growth Analysis

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Welcome to the Strategic EV Market Intelligence Dashboard

This project evaluates India's Electric Vehicle (EV) ecosystem to identify growth opportunities, market expansion strategies, and manufacturing feasibility for AtliQ Motors.
Using EV sales data, CAGR trends, government subsidies, charging infrastructure insights, and revenue projections, this dashboard highlights key patterns in adoption rates, policy impact, and long-term scalability.

The analysis is structured into three key sections:

- 🔹 **Problem Statement** — Understanding the EV market landscape, challenges, and business objectives
- 🔹 **Primary Analysis** — EV sales trends, penetration rates, CAGR, revenue growth, and infrastructure correlation
- 🔹 **Strategic Recommendations** — Manufacturing location analysis, market entry strategy, and expansion roadmap

Use the navigation buttons below to explore each section of the report.

🎯 **Problem Statement**

📌 **Primary Task**

🧠 **Secondary Task**

📌 Problem Statement

🚨 Market Opportunity & Urgency

India is rapidly emerging as one of the fastest-growing electric vehicle markets due to rising fuel prices, environmental concerns, and strong government incentives.
While EV adoption is accelerating, competition is intensifying, making data-driven expansion decisions critical for long-term success.

🚗 Business Challenge

AtliQ Motors, a leading automotive giant from the USA, holds a 25% market share in the EV and hybrid segment in North America.
However, its presence in India is currently below 2%, creating both a challenge and a significant growth opportunity.
Before launching bestselling EV/Hybrid models in India, the company needs to clearly understand:

- State-wise EV sales and penetration trends
- 2-wheeler vs 4-wheeler adoption patterns
- Impact of government incentives and subsidies
- Charging infrastructure readiness
- Revenue growth potential and CAGR trends

A comprehensive market study is essential to reduce risk and ensure a successful market entry strategy.

📊 Analysis Objectives

The leadership team has assigned the data analytics division to conduct a detailed EV market assessment.

The key objectives include:
1️⃣ Market Growth Analysis — Evaluate CAGR, sales trends, and penetration rates
2️⃣ Segment & Revenue Study — Analyze performance of 2-wheelers and 4-wheelers
3️⃣ Policy & Infrastructure Impact — Measure subsidy influence and charging station correlation
4️⃣ Strategic Recommendation — Identify ideal manufacturing location and expansion roadmap
The final deliverable is a self-explanatory dashboard providing actionable insights for strategic decision-making.

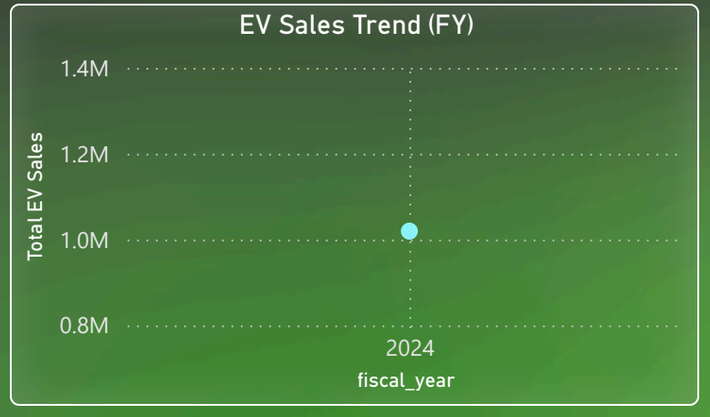
👤 Dashboard by **Ishwari Attarde** . 📧 ishwariattarde02@gmail.com

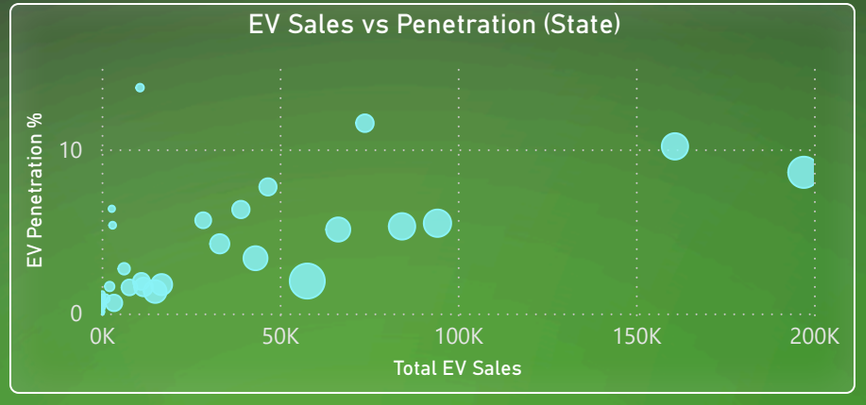
INDIA EV MARKET OVERVIEW

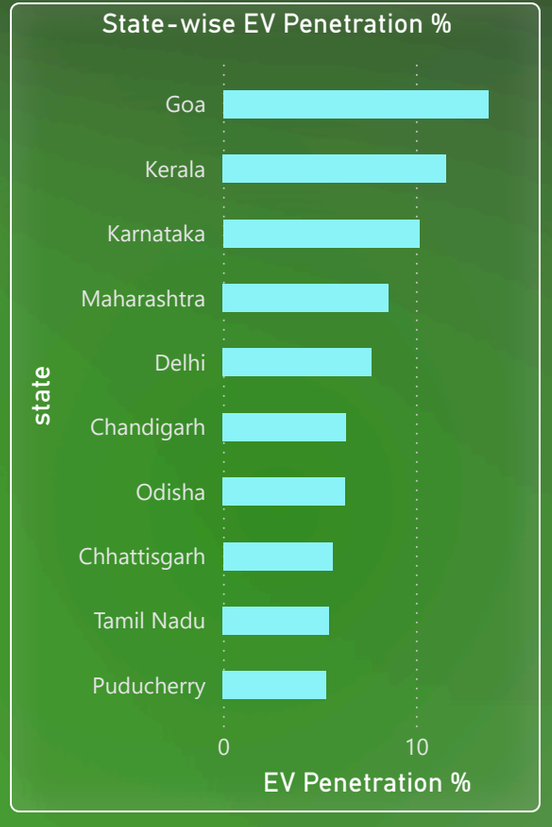
1M	4.81%	31.5%	93.9%
Total EV Sales	EV Penetration %1	YoY Growth %	EV CAGR %

2023 | 2024

2-Wheelers | 4-Wheelers

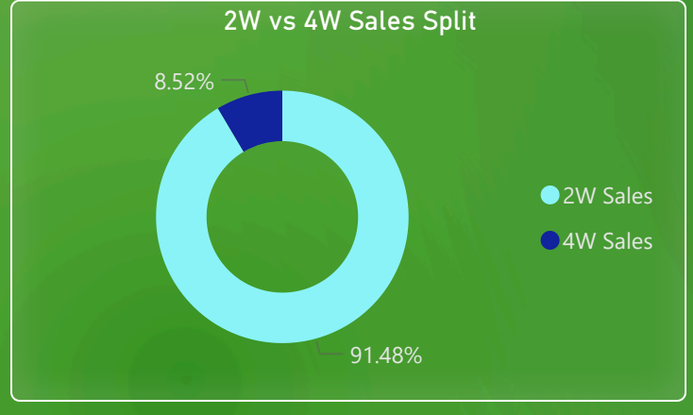






State EV Performance

state	Total EV Sales	EV Penetration %	EV CAGR %
Andaman & Nicobar	2	0.30	-1.00
Andaman & Nicobar Island	33	0.50	0.22
Andhra Pradesh	33183	4.24	0.54
Arunachal Pradesh	31	0.11	-1.00
Assam	3497	0.64	1.19
Bihar	15069	1.33	0.77
Chandigarh	2877	6.37	1.65



COMPETITION LANDSCAPE

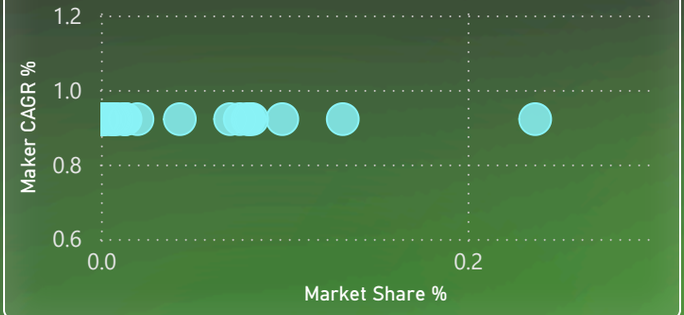
26	100.0%	92.2%
Total Makers	Market Share %	Maker CAGR %

fiscal_year: All

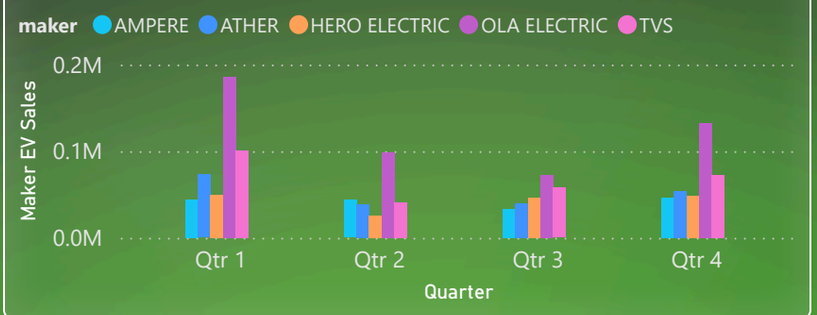
maker: All

2-Wheelers | 4-Wheelers

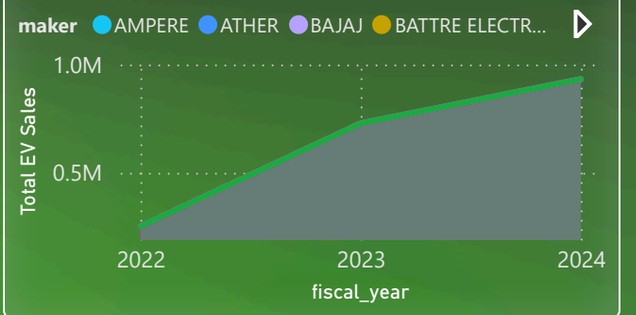
Maker Share vs CAGR



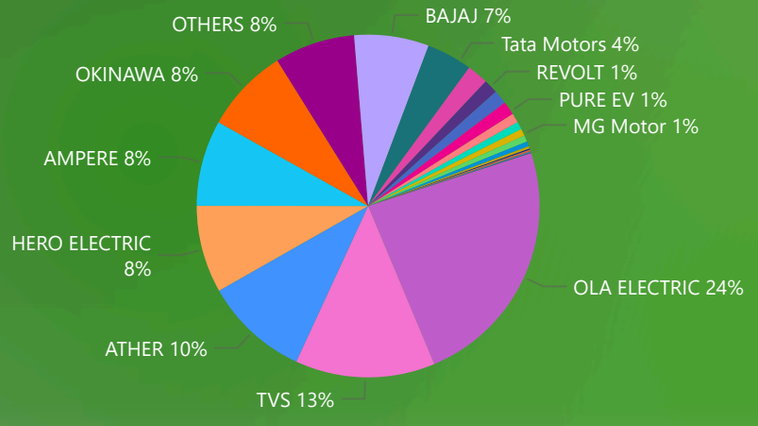
Quarterly Sales by Maker



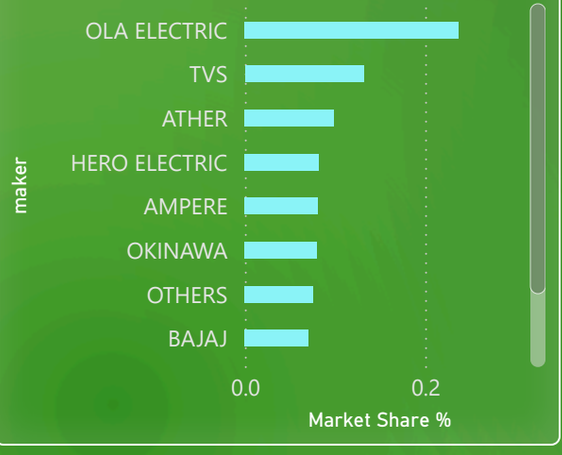
FY Sales by Maker



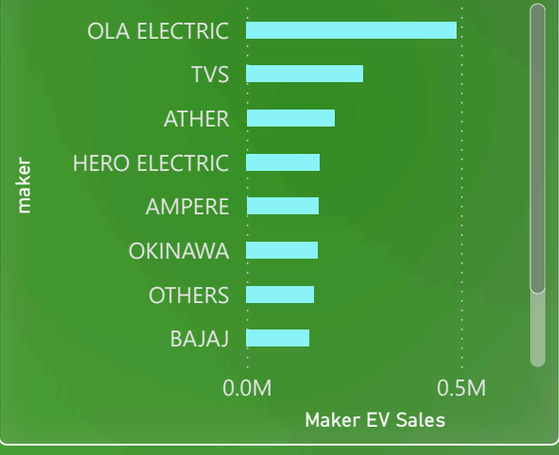
Maker Market Share



Share % by Maker

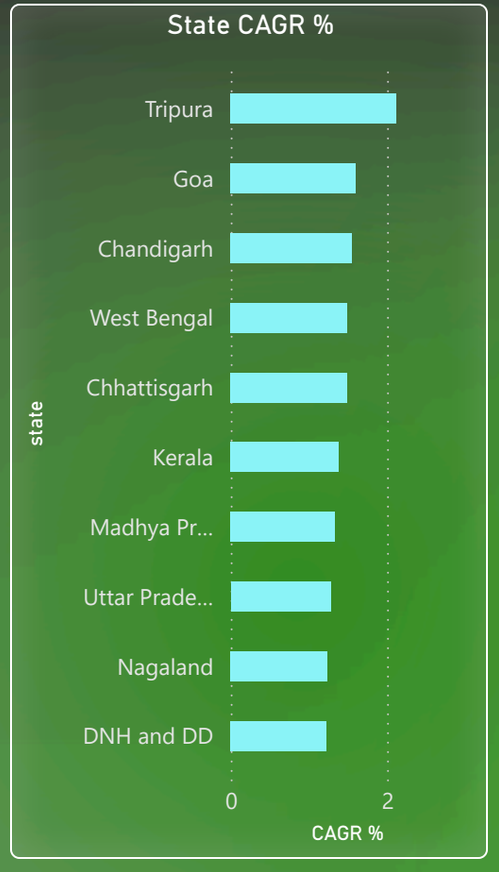


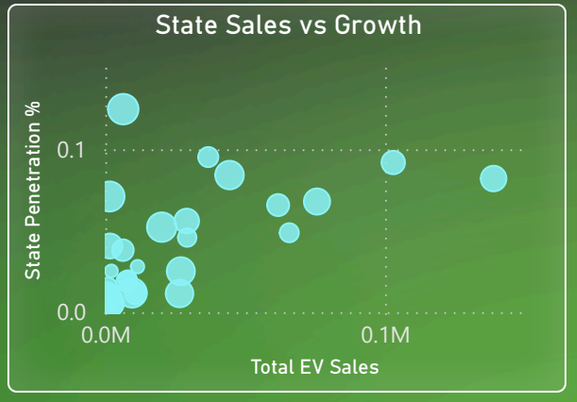
Top 10 EV Makers



STATE OPPORTUNITY ANALYSIS

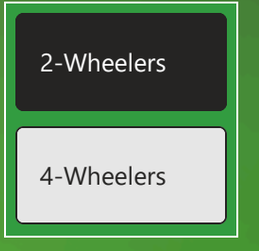
4.54%	81.1	16	92.2%
State Penetration %	Opportunity Score	Rank by CAGR	CAGR %





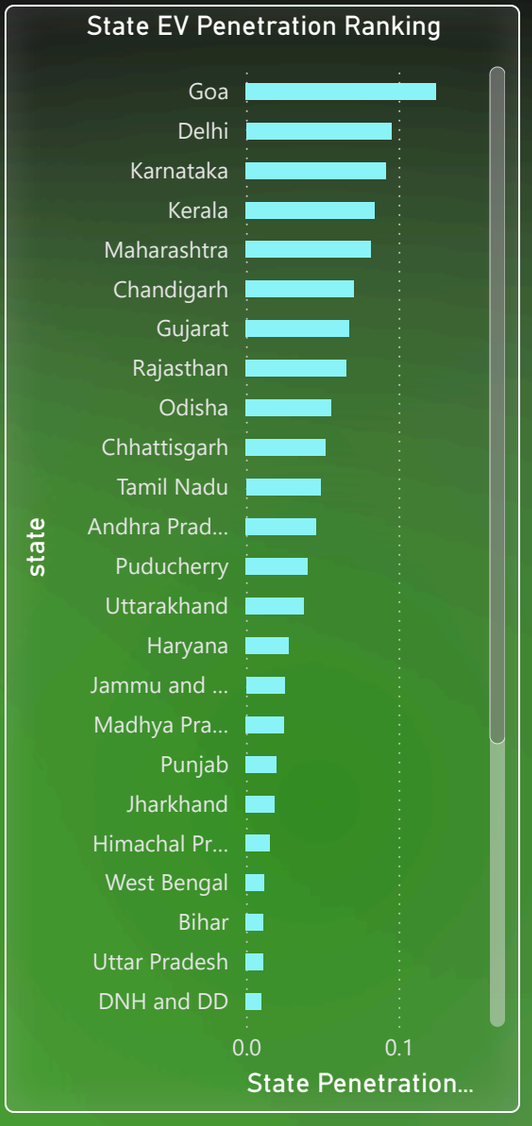






State Sales & Growth

state	EV Sales FY 2022	EV Sales FY 2024	Sales Growth %	CAGR %
Andaman & Nicobar		2		
Andaman & Nicobar Island	0	18		
Andhra Pradesh	13465	31353	132.85	0.53
Arunachal Pradesh	0	3		
Assam	719	3168	340.61	1.10
Bihar	4750	14366	202.44	0.74
Chandigarh	286	1857	549.30	1.55



FUTURE FORECAST & SEASONALITY

21K
Peak Month Sales

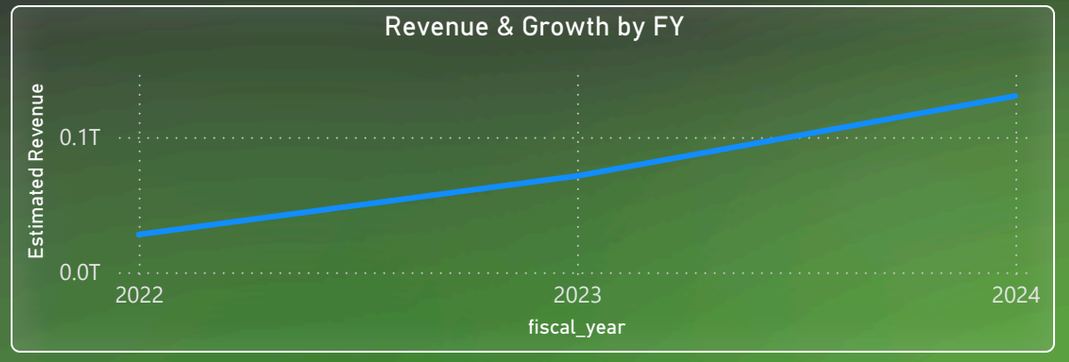
9K
Lowest Month Sales

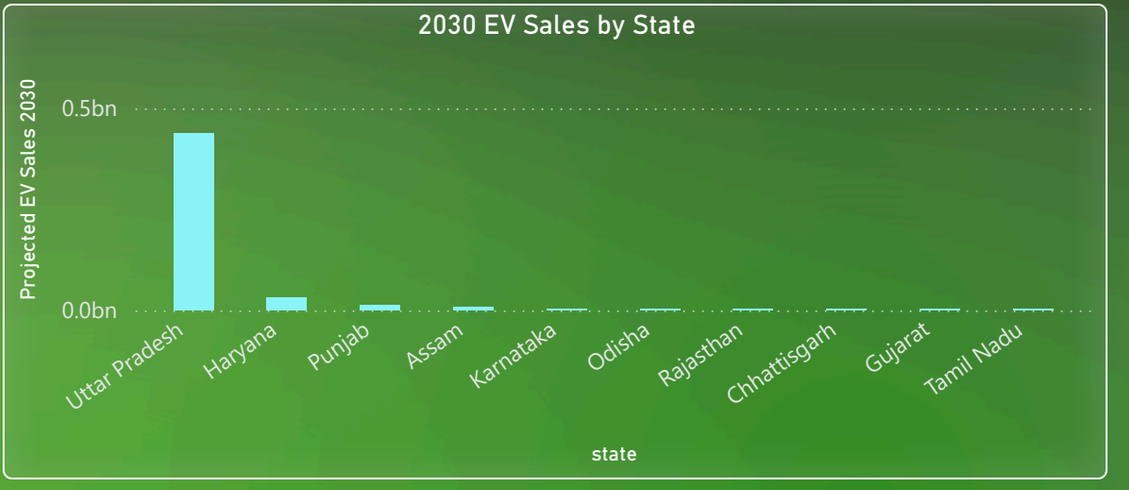
8.90M
Projected EV Sales 2030

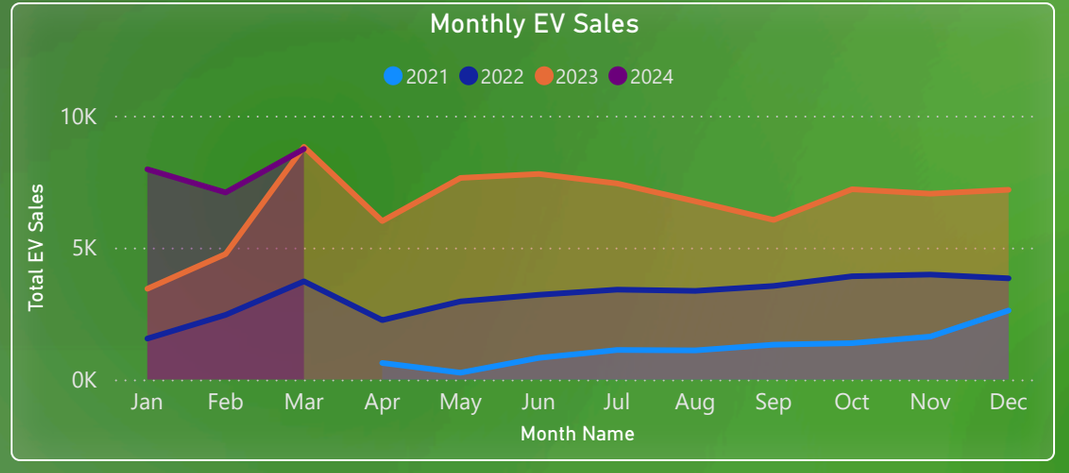
2-Wheelers | 4-Wheelers

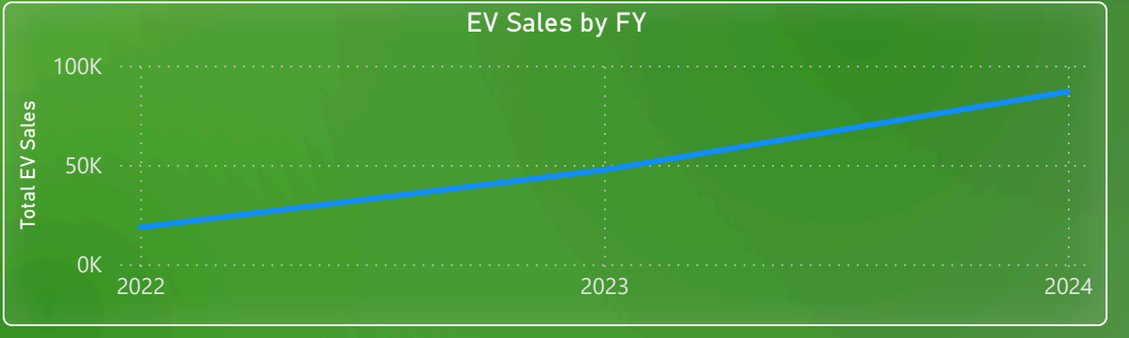
state
All

fiscal_year
2022
2024









Strategic Business Intelligence

83.10
Opportunity Score

54.2M
Projected EV Sales 2030

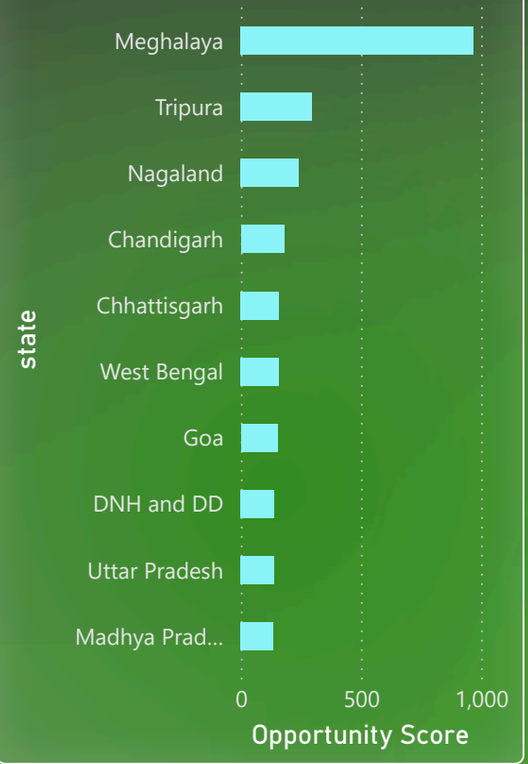
2-Wheelers
Top Segment

2-Wheelers | 4-Wheelers

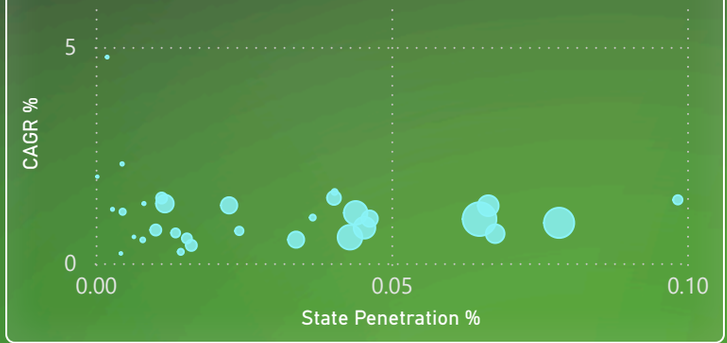
fiscal_year: 2022 – 2024

state: All

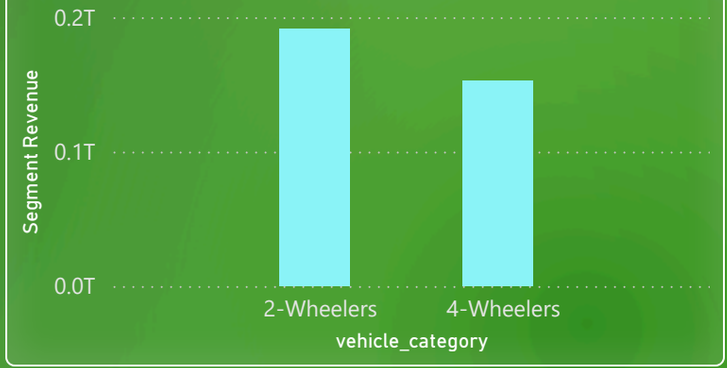
State Opportunity Score



State Penetration, CAGR & Sales



State Sales & Growth



State EV Metrics

state	CAGR %	Penetration Growth %	Total EV Sales
Meghalaya	4.77	0.00	177
Tripura	2.30	0.01	562
Nagaland	2.00	0.00	13
Chandigarh	1.65	0.05	5279
Chhattisgarh	1.51	0.05	53804
West Bengal	1.51	0.01	30560
Goa	1.46	0.10	19684
DNH and DD	1.38	0.01	355
Uttar Pradesh	1.38	0.02	95203
Madhya Pradesh	1.34	0.03	78979
Kerala	1.33	0.10	137060
Manipur	1.24	0.01	299
Assam	1.19	0.00	6418
Gujarat	1.16	0.04	181389
Puducherry	1.05	0.04	5536
Odisha	1.03	0.04	78267
Maharashtra	1.02	0.06	396045
Karnataka	0.93	0.06	312995
Rajasthan	0.82	0.03	150366

Preliminary Research Questions

1. List the top 3 and bottom 3 makers for the fiscal years 2023 and 2024 in terms of the number of 2-wheelers sold. Q 1

2. Identify the top 5 states with the highest penetration rate in 2-wheeler and 4-wheeler EV sales in FY 2024. Q 2

3. List the states with negative penetration (decline) in EV sales from 2022 to 2024? Q 3

4. What are the quarterly trends based on sales volume for the top 5 EV makers (4-wheelers) from 2022 to 2024? Q 4

5. How do the EV sales and penetration rates in Delhi compare to Karnataka for 2024? Q 5

6. List down the compounded annual growth rate (CAGR) in 4-wheeler units for the top 5 makers from 2022 to 2024. Q 6

7. List down the top 10 states that had the highest compounded annual growth rate (CAGR) from 2022 to 2024 in total vehicles sold. Q 7

8. What are the peak and low season months for EV sales based on the data from 2022 to 2024? Q 8

9. What is the projected number of EV sales (including 2-wheelers and 4 wheelers) for the top 10 states by penetration rate in 2030, based on the compounded annual growth rate (CAGR) from previous years? Q 9

10. Estimate the revenue growth rate of 4-wheeler and 2-wheelers EVs in India for 2022 vs 2024 and 2023 vs 2024, assuming an average unit price. Q 10

Top 3 and Bottom 3 Makers (2-Wheelers) – FY 2023 & 2024

2023 2024

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======== FY 2023 ========

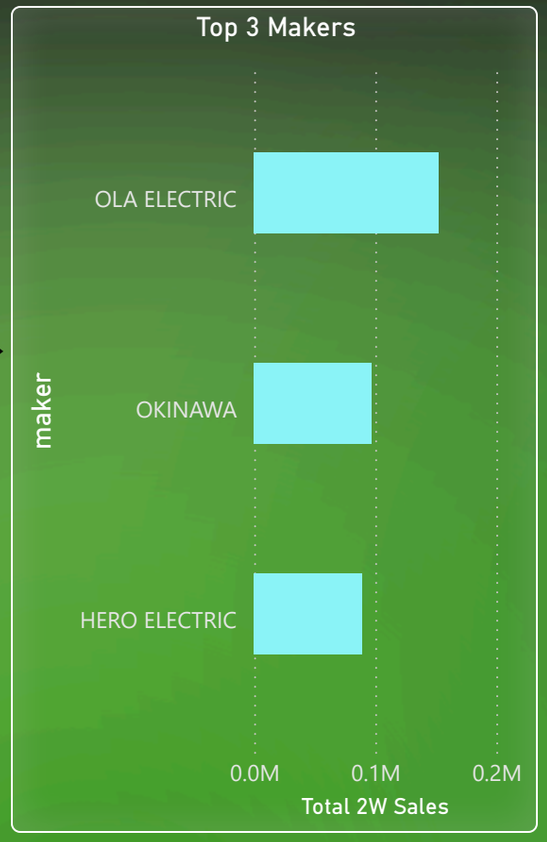
Top 3 Makers (2W):
    fiscal_year          maker  electric_vehicles_sold
20         2023   OLA ELECTRIC                  152583
19         2023       OKINAWA                   96945
16         2023  HERO ELECTRIC                  88993

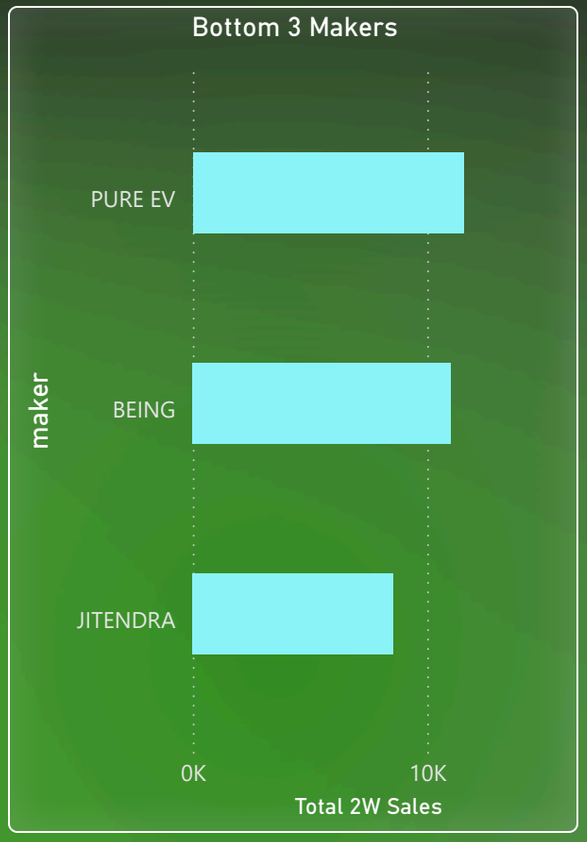
Bottom 3 Makers (2W):
    fiscal_year     maker  electric_vehicles_sold
22         2023   PURE EV                   11556
15         2023     BEING                   11018
17         2023  JITENDRA                    8563

======== FY 2024 ========

Top 3 Makers (2W):
    fiscal_year         maker  electric_vehicles_sold
34         2024  OLA ELECTRIC                  322489
37         2024           TVS                  180743
26         2024         ATHER                  107552

Bottom 3 Makers (2W):
    fiscal_year            maker  electric_vehicles_sold
31         2024    KINETIC GREEN                    9585
36         2024           REVOLT                    7254
28         2024  BATTRE ELECTRIC                    4841
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Top 5 States – Highest Penetration (2W & 4W) – FY 2024

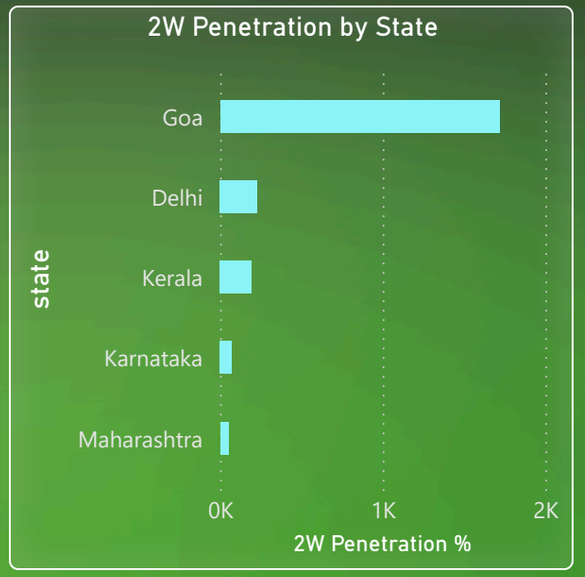
2024

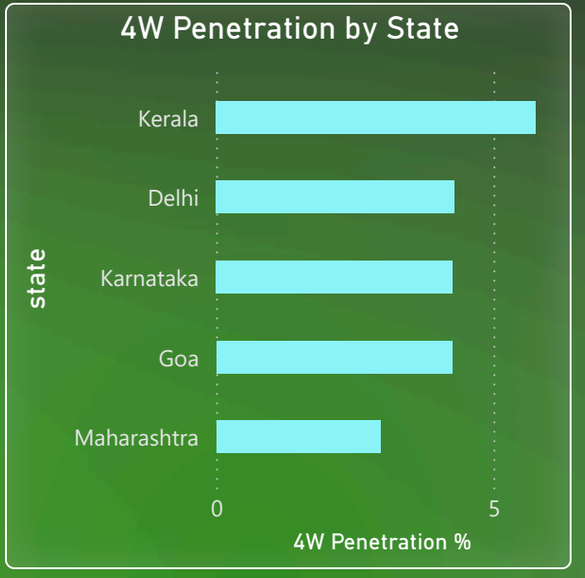
🔹 Top 5 States – 2W Penetration (FY 2024)

	state	electric_vehicles_sold	total_vehicles_sold	Penetration_%
10	Goa	9768	54290	17.992264
17	Kerala	64769	478887	13.524903
16	Karnataka	148111	1279767	11.573279
20	Maharashtra	183052	1817343	10.072507
9	Delhi	38094	405218	9.400866

🔹 Top 5 States – 4W Penetration (FY 2024)

	state	electric_vehicles_sold	total_vehicles_sold	Penetration_%
17	Kerala	9169	159227	5.758445
6	Chandigarh	1020	22651	4.503112
9	Delhi	8630	201130	4.290757
16	Karnataka	12878	302221	4.261120
10	Goa	1031	24234	4.254353





"In FY 2024, the top 5 states in 2W and 4W penetration demonstrate strong EV adoption maturity and represent high-priority expansion markets."

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States with Negative Penetration Growth (2022 → 2024)

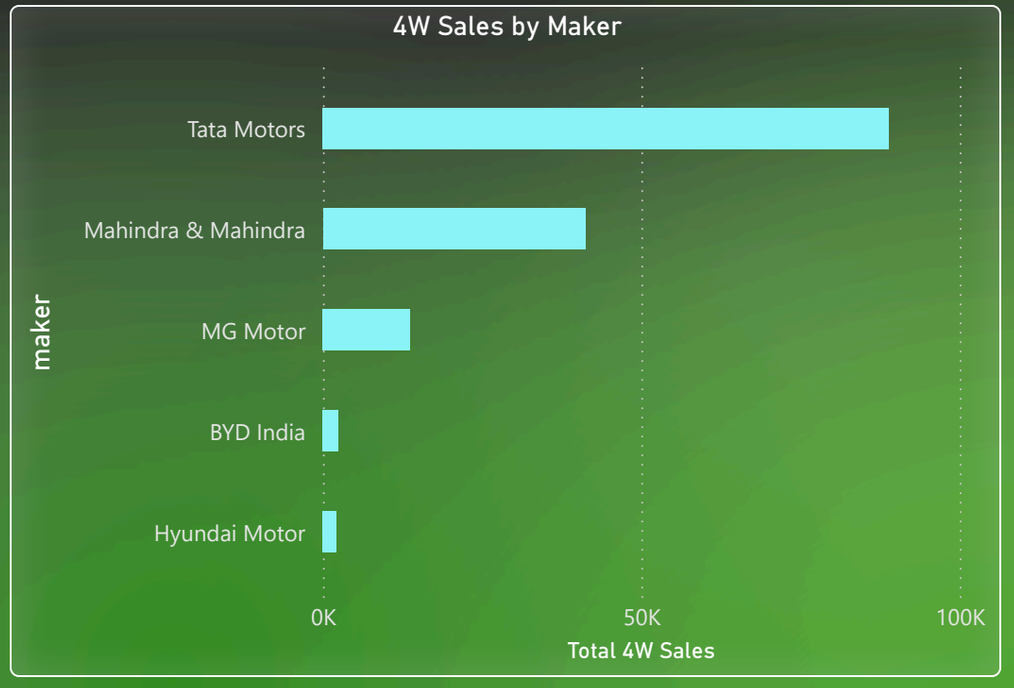
state	Penetration FY 2022	Penetration FY 2024	Penetration Change
Andaman & Nicobar		0.00	0.00
Andaman & Nicobar Island	0.00	0.01	0.00
Andhra Pradesh	0.02	0.04	0.02
Arunachal Pradesh	0.00	0.00	0.00
Assam	0.00	0.01	0.00
Bihar	0.01	0.01	0.01
Chandigarh	0.01	0.06	0.05
Chhattisgarh	0.01	0.06	0.05
Delhi	0.04	0.08	0.04
DNH and DD	0.00	0.01	0.01
Goa	0.04	0.14	0.10
Gujarat	0.02	0.05	0.04
Haryana	0.01	0.02	0.00
Himachal Pradesh	0.00	0.01	0.00
Jammu and Kashmir	0.01	0.02	0.01
Jharkhand	0.01	0.02	0.01
Karnataka	0.04	0.10	0.06
Kerala	0.02	0.12	0.10
Ladakh	0.00	0.01	0.01

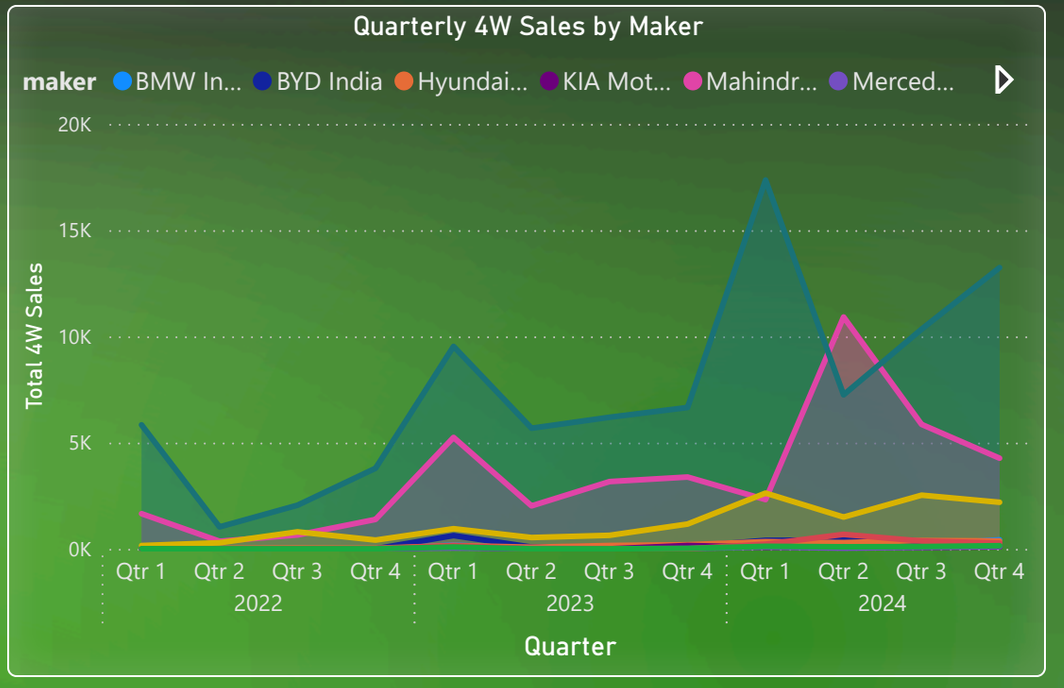
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"The listed states show a decline in EV penetration from 2022 to 2024, indicating slowing adoption momentum and potential policy or infrastructure gaps."

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Quarterly Trends – Top 5 4W Makers (2022–2024)





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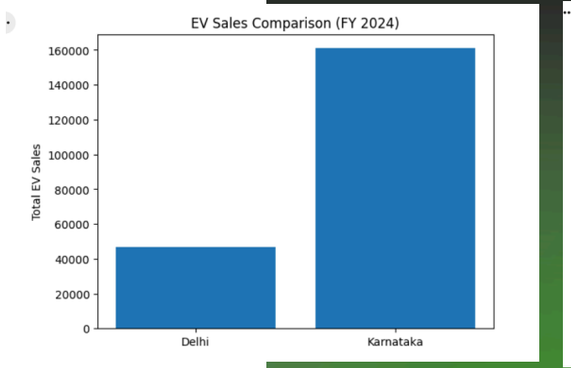
2M
Total EV Sales

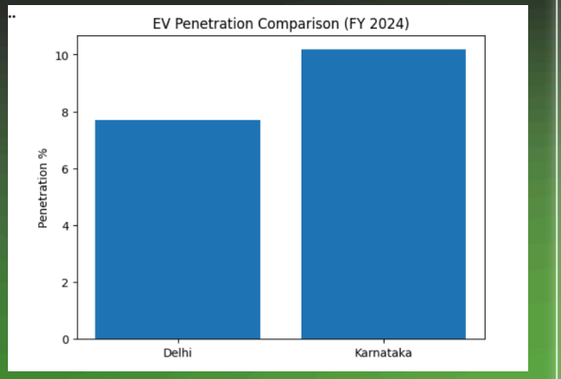
47K
Delhi EV Sales (2024)

161K
Karnataka EV Sales (2024)

10.18%
Karnataka Penetration %...

7.71%
Delhi Penetration % (2024)





State EV Sales & Penetration

state	Total EV Sales	EV Penetration %
Karnataka	312995	7.84
Delhi	107312	6.76
Total	**420307**	**7.53**

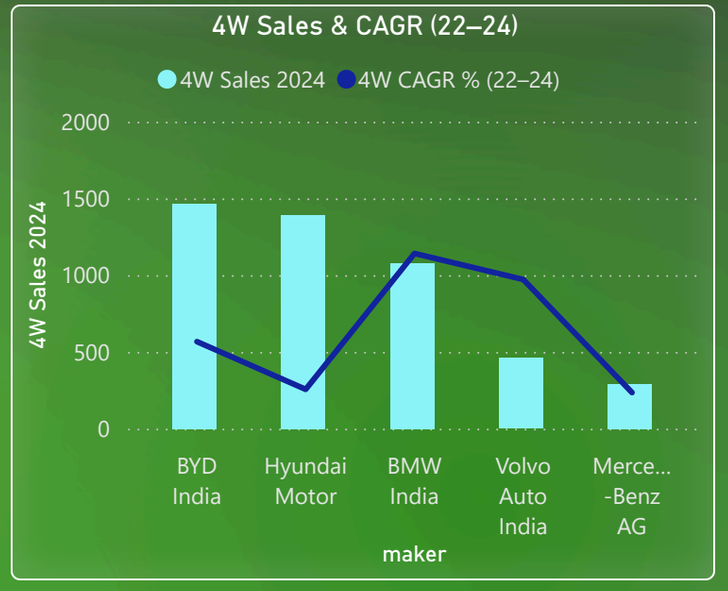
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CAGR of 4-Wheeler Units (2022–2024) – Top 5 Makers

CAGR (2022–2024) for Top 5 4W EV Makers:

	maker	Sales_2022	Sales_2024	CAGR_%
0	BYD India	33	1466	566.515134
1	Hyundai Motor	110	1390	255.476633
2	MG Motor	1647	8829	131.530899
3	Mahindra & Mahindra	4042	23346	140.330055
4	Tata Motors	12708	48181	94.714952

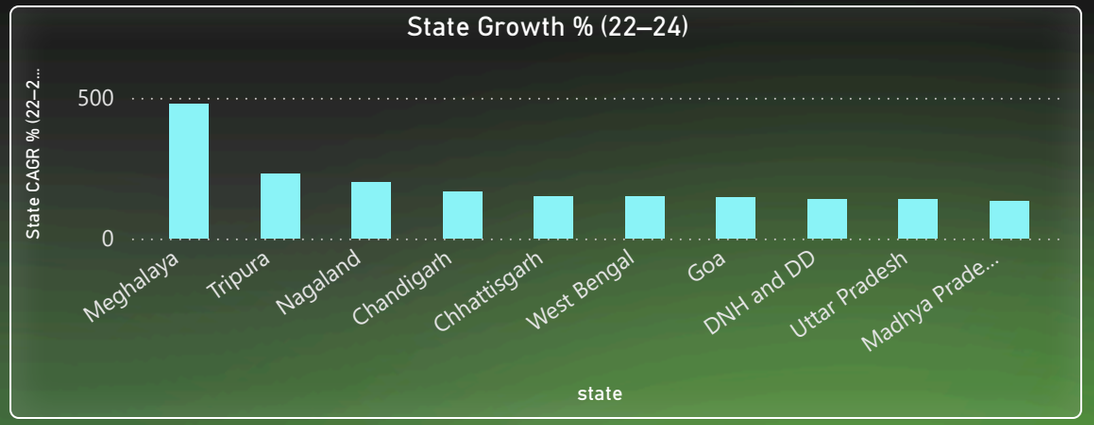


"Among the Top 5 4W EV makers, CAGR analysis (2022–2024) reveals the fastest-growing brands, highlighting strong competitive momentum and long-term market expansion."

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Top 10 States by CAGR (2022–2024)



Meghalaya
Highest CAGR State

192.79
Average CAGR (Top 10)

Top 10 States by CAGR (2022–2024) - Total EV Sales

	state	Sales_2022	Sales_2024	CAGR_%
0	Meghalaya	4.0	133.0	476.628130
1	Tripura	28.0	304.0	229.501788
2	Nagaland	1.0	9.0	200.000000
3	Chandigarh	411.0	2877.0	164.575131
4	Chhattisgarh	4534.0	28540.0	150.891661
5	West Bengal	2685.0	16864.0	150.615629
6	Goa	1778.0	10799.0	146.448337
7	DNH and DD	35.0	198.0	137.847490
8	Uttar Pradesh	10222.0	57758.0	137.704900
9	Madhya Pradesh	7916.0	43223.0	133.670862

"The top 10 states with the highest CAGR (2022–2024) indicate accelerating EV adoption momentum, reflecting regional infrastructure expansion and policy effectiveness."

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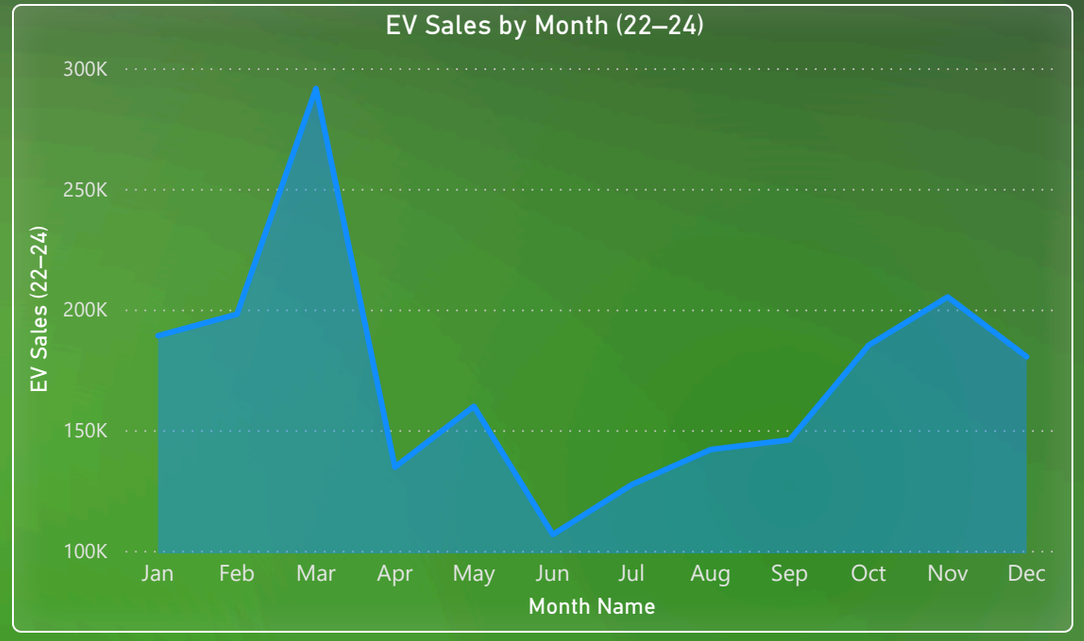
Peak and Low Season Months (2022–2024)

292K
peak Month Sales

107K
Lowest Month Sales

📊 Full Monthly Sales Summary:

	Month	Month_Name	electric_vehicles_sold
0	1	January	189099
1	2	February	198049
2	3	March	291587
3	4	April	128342
4	5	May	158370
5	6	June	101222
6	7	July	111632
7	8	August	124808
8	9	September	126621
9	10	October	162995
10	11	November	179037
11	12	December	151160



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"The seasonal trend from 2022–2024 shows that EV sales peak during [Peak Month] and dip during [Low Month], indicating strong seasonal demand cycles."

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Projected EV Sales by 2030 (Top 10 States by Penetration)

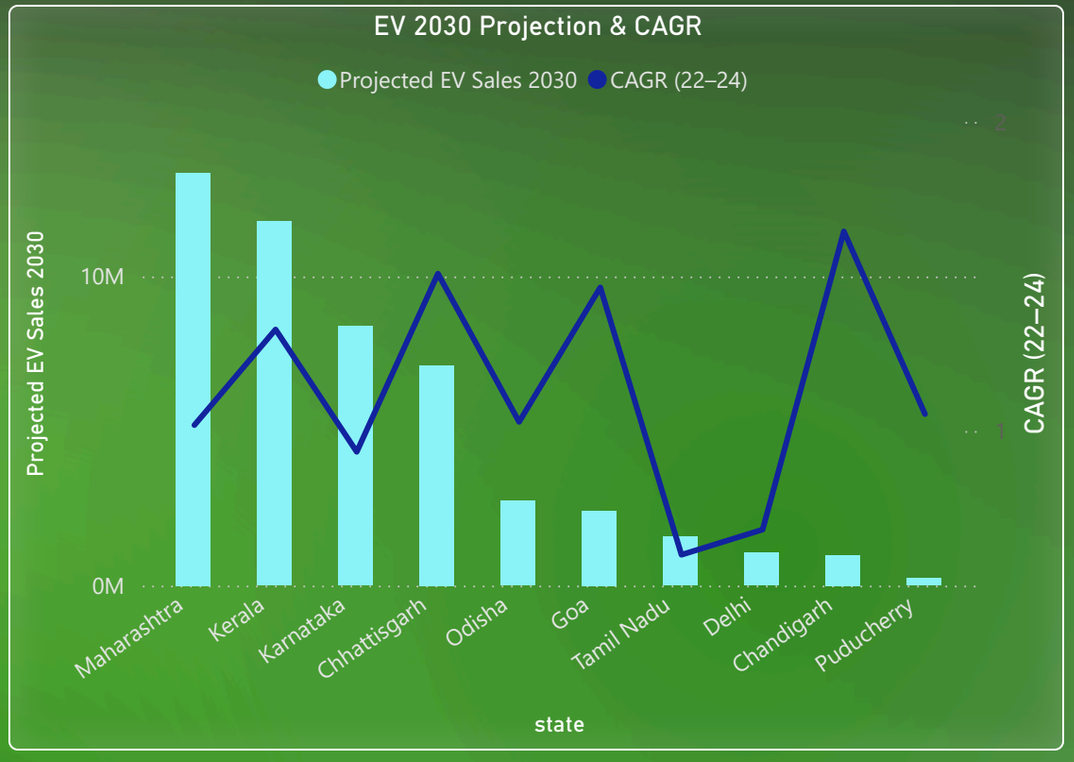
Maharashtra
Highest Projected State

49.64M
Total Projected Sales (Top 10)

Projected EV Sales for 2030 (Top 10 States by 2024 Penetration Rate)

fiscal_year	state	Sales_2022	Sales_2024	CAGR
0	Goa	1778.0	10799.0	1.464483
1	Kerala	13639.0	73938.0	1.328320
2	Karnataka	43111.0	160989.0	0.932431
3	Maharashtra	48374.0	197169.0	1.018893
4	Delhi	16535.0	46724.0	0.681001
5	Chandigarh	411.0	2877.0	1.645751
6	Odisha	9498.0	39118.0	1.029421
7	Chhattisgarh	4534.0	28540.0	1.508917
8	Tamil Nadu	36863.0	94314.0	0.599531
9	Puducherry	734.0	3098.0	1.054436

fiscal_year	Projected_2030_Sales
0	2.419574e+06
1	1.177940e+07
2	8.383406e+06
3	1.335115e+07
4	1.054259e+06
5	9.868110e+05
6	2.732814e+06
7	7.118219e+06
8	1.579547e+06
9	2.329365e+05



"Using CAGR from 2022–2024, projected 2030 EV sales indicate exponential growth potential in high-penetration states, suggesting strong long-term adoption momentum."

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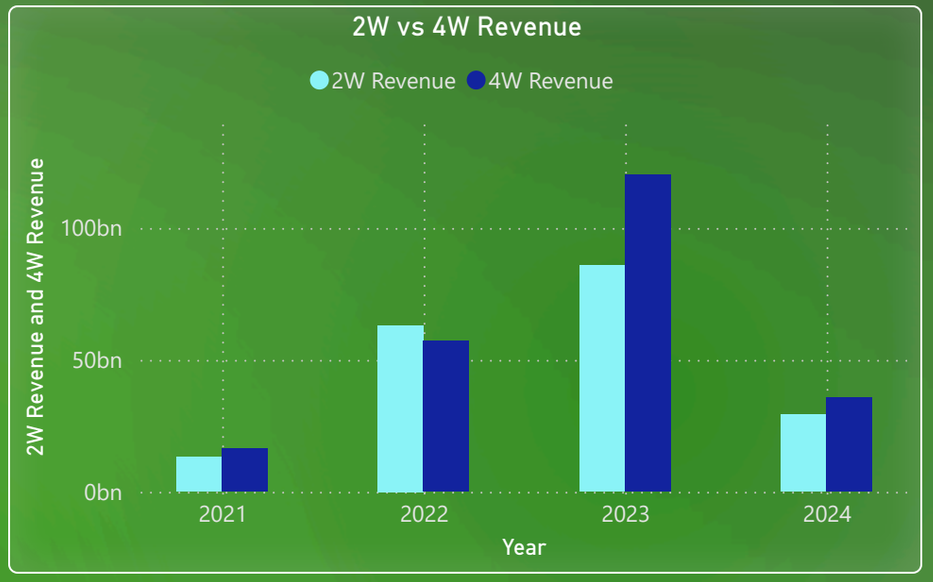
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Revenue Growth Rate (Estimated using Average Unit Price)

171.05
Growth % 22-23

55.31
Growth % 23-24

fiscal_year	Revenue_2022	Revenue_2023	Revenue_2024	Growth_%_22_vs_24
vehicle_category				
2-Wheelers	21468705000	61871755000	79278820000	269.276209
4-Wheelers	27865500000	71197500000	130351500000	367.788125



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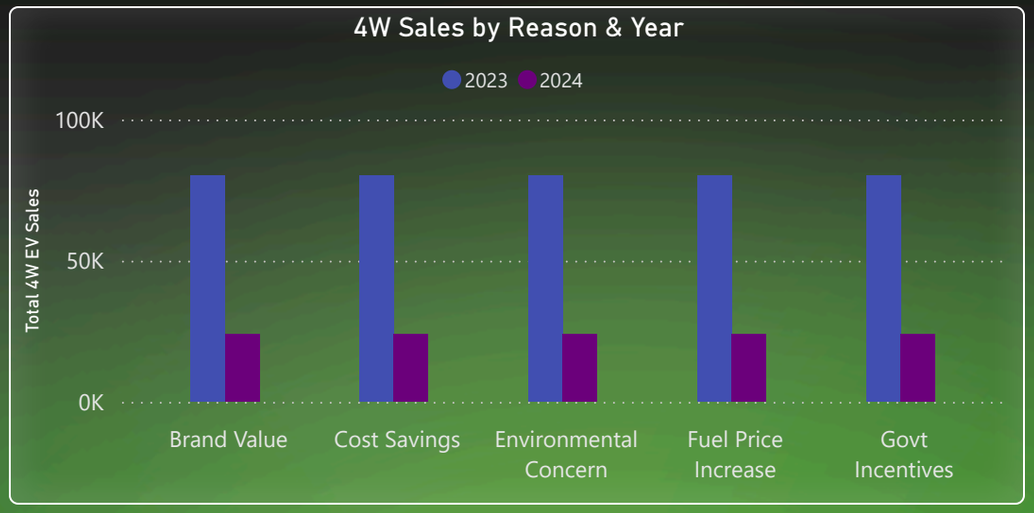
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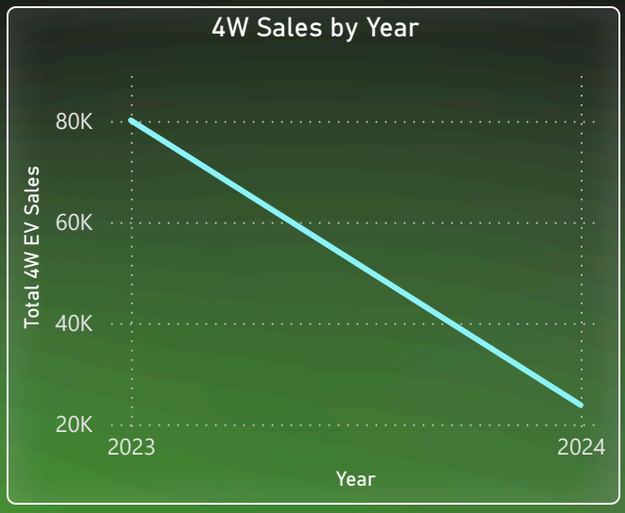
Secondary Research Questions

1. What are the primary reasons for customers choosing 4-wheeler EVs in 2023 and 2024 (cost savings, environmental concerns, government incentives)?	Q 1
2. How do government incentives and subsidies impact the adoption rates of 2-wheelers and 4-wheelers? Which states in India provided most subsidies?	Q 2
3. How does the availability of charging stations infrastructure correlate with the EV sales and penetration rates in the top 5 states?	Q 3
4. Who should be the brand ambassador if AtliQ Motors launches their EV/Hybrid vehicles in India and why?	Q 4
5. Which state of India is ideal to start the manufacturing unit? (Based on subsidies provided, ease of doing business, stability in governance etc.)	Q 5
6. Your top 3 recommendations for AtliQ Motors.	Q 6

Primary Reasons for Choosing 4-Wheeler EVs (2023–2024)

83.08
4W Revenue Growth % (23-24)



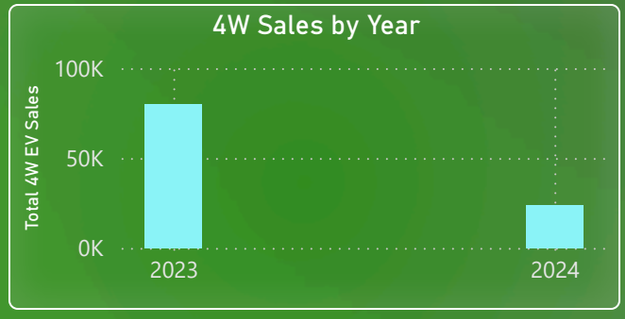


Customer Reason Distribution (2023 vs 2024)

	Reason	2023 %	2024 %
0	Cost Savings	35	32
1	Environmental Concern	28	30
2	Government Incentives	22	20

✅ 4-Wheeler EV Sales Trend (2023–2024)

	year	electric_vehicles_sold	Growth_Rate_%
0	2023	80121	NaN
1	2024	23794	-70.302418



In 2023–2024, customers primarily choose 4W EVs due to long-term cost savings and environmental awareness, while dependence on government incentives is gradually reducing — indicating market maturity.

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Government Incentives & Adoption Impact

2024

Top States Providing Highest Subsidies (2024):

	state	vehicle_category	electric_vehicles_sold	revenue	subsidy_amount	adoption_rate_%
40	Maharashtra	2-Wheelers	183052	15559420000	2.333913e+09	10.157240
41	Maharashtra	4-Wheelers	14117	21175500000	2.117550e+09	3.004033
33	Karnataka	4-Wheelers	12878	19317000000	1.931700e+09	4.277313
32	Karnataka	2-Wheelers	148111	12589435000	1.888415e+09	11.510878
35	Kerala	4-Wheelers	9169	13753500000	1.375350e+09	42.309117
19	Delhi	4-Wheelers	8630	12945000000	1.294500e+09	4.263585
60	Tamil Nadu	2-Wheelers	87405	7429425000	1.114414e+09	6.116695
23	Gujarat	4-Wheelers	7257	10885500000	1.088550e+09	2.102633
61	Tamil Nadu	4-Wheelers	6909	10363500000	1.036350e+09	2.595815
22	Gujarat	2-Wheelers	77102	6553670000	9.830505e+08	6.442696

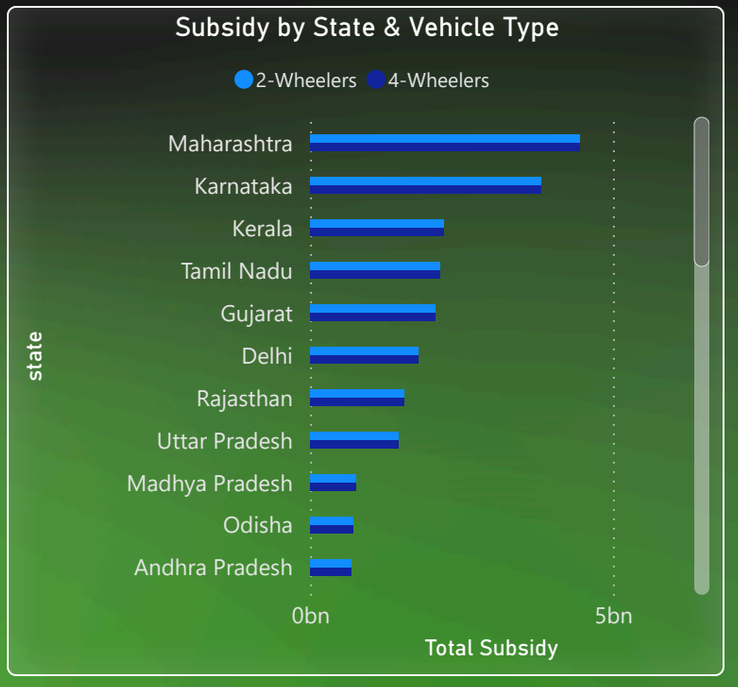
Impact of Subsidies on 2W vs 4W (2024):

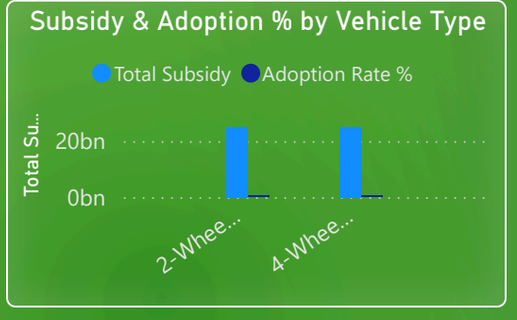
	vehicle_category	electric_vehicles_sold	revenue	subsidy_amount	adoption_rate_%
0	2-Wheelers	932692	79278820000	1.189182e+10	4.286377
1	4-Wheelers	86901	130351500000	1.303515e+10	2.712491

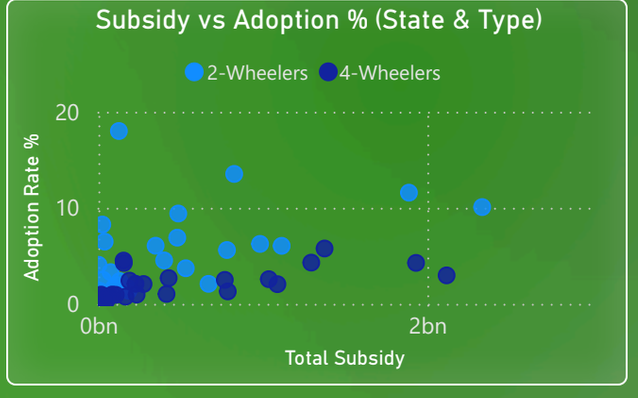
4.81%
Adoption Rate %1

24.9bn
Total Subsidy

210bn
Total Revenue1







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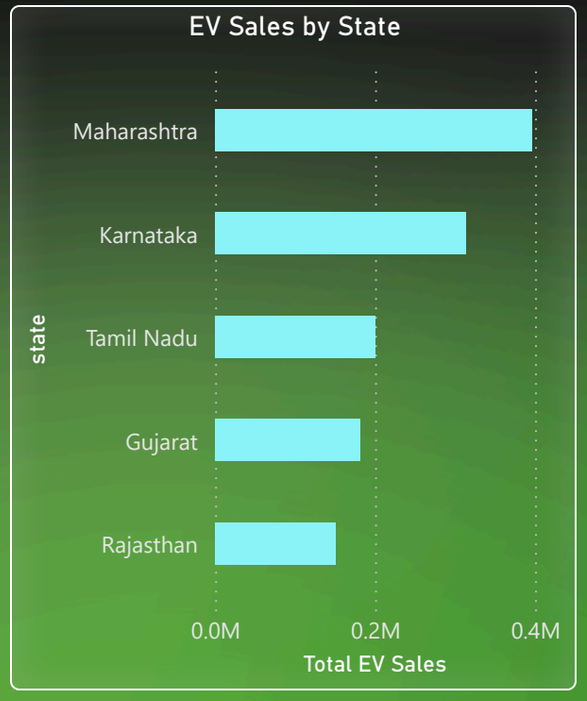
Charging Infrastructure & EV Sales Correlation

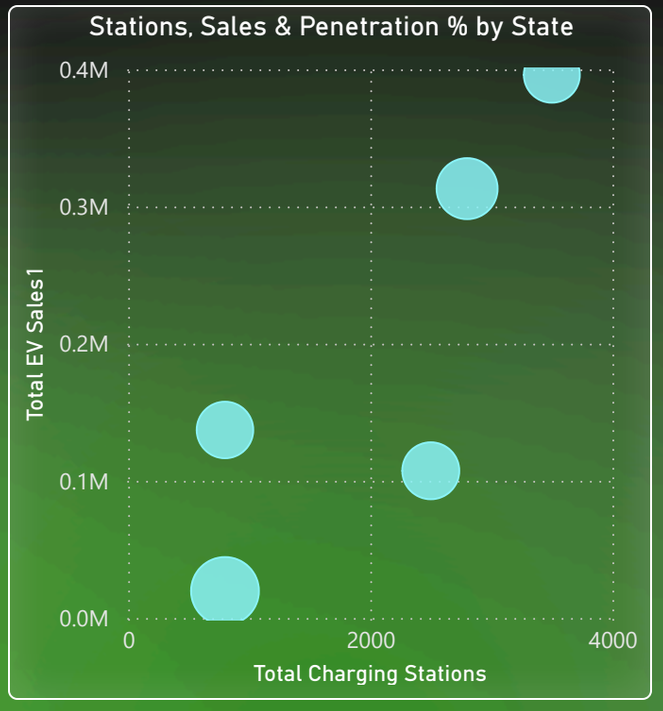
Stations, Sales & Penetration % by State

state	Total Charging Stations	Total EV Sales1	Penetration Rate %
Delhi	2500	107312	6.76
Goa	800	19684	9.84
Karnataka	2800	312995	7.84
Kerala	800	137060	6.64
Maharashtra	3500	396045	6.49

Top 5 States Summary:

	state	electric_vehicles_sold	penetration_rate_%	charging_stations
0	Gujarat	84359	4.272664	2000.0
1	Karnataka	160989	7.894095	2800.0
2	Kerala	73938	27.959755	800.0
3	Maharashtra	197169	6.580637	3500.0
4	Tamil Nadu	94314	4.356255	2200.0





2.56
Avg Penetration %

2M
Total EV Sales1

4K
Total Charging Stations

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Brand Ambassador Recommendation

🏏 Hardik Pandya

Hardik Pandya would be a strong and trendy choice as the brand ambassador for AtliQ Motors' EV/Hybrid launch in India. He represents **confidence, ambition, and a modern lifestyle**, which aligns perfectly with the positioning of electric vehicles as innovative and future-focused mobility solutions. His strong influence among young urban consumers makes him ideal for targeting the fast-growing EV buyer segment.

🎯 Why He Fits the Brand:

- **Youth Appeal:** Connects strongly with millennials and Gen Z, the key EV adoption group.
- **Pan-India Popularity:** Recognized across metro and Tier-1 cities.
- **Aspirational Image:** Represents success, performance, and bold decision-making.
- **Strong Digital Presence:** High engagement on social media helps boost brand visibility and campaign reach.

🚀 Strategic Impact:

Partnering with Hardik Pandya can position AtliQ Motors as:

- A bold and performance-driven EV brand
- A modern, tech-forward mobility solution
- A lifestyle upgrade rather than just a vehicle purchase

This alignment can significantly strengthen brand recall and accelerate market entry success in India.

Ideal State for Manufacturing

Maharashtra
Best Manufacturing State

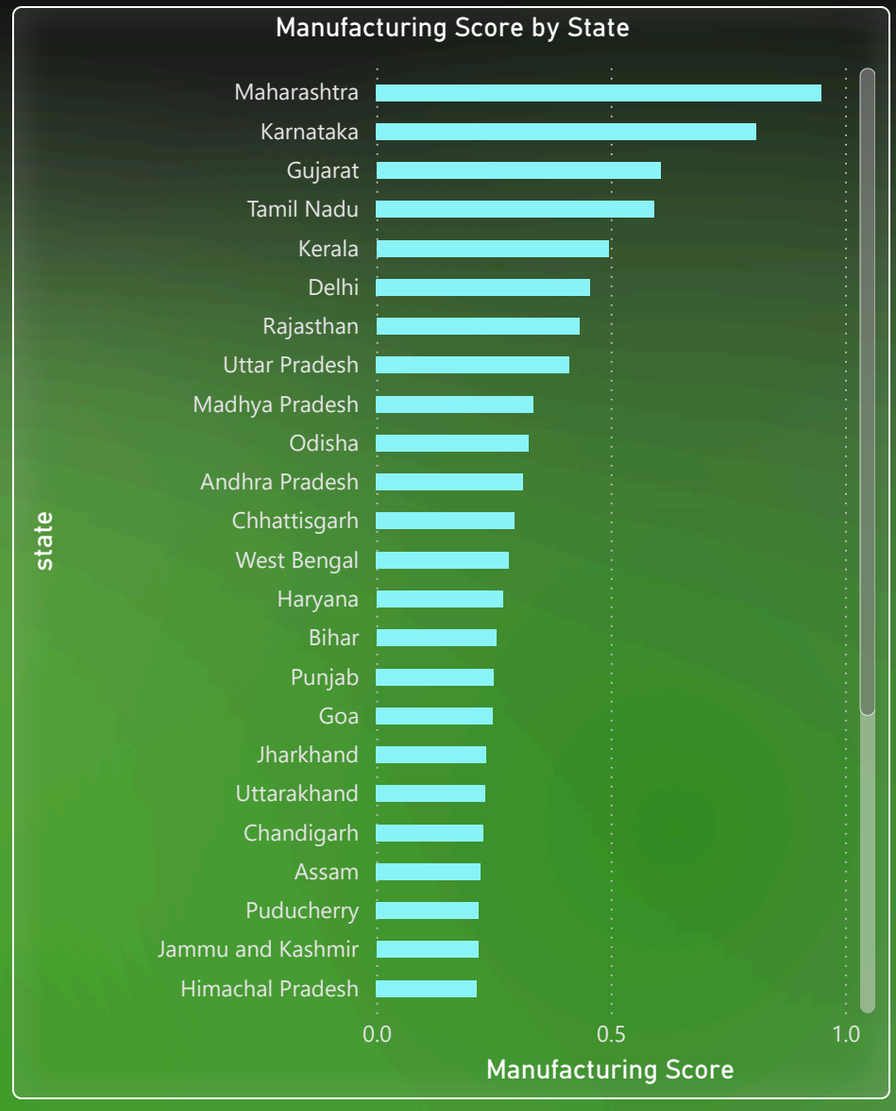
Meghalaya
Top Ranked State

4.45bn
Top State Subsidy

Year
2024

0.95
Highest Manufacturing Score

197K
Top State EV Sales



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Top 3 Recommendations

🏭 1️⃣ Establish Manufacturing in a High-Subsidy, Business-Friendly State
📌 **Recommendation:** Set up the manufacturing unit in a state offering strong EV subsidies, industrial infrastructure, and policy stability.
This will reduce capital investment, optimize operational costs, and ensure long-term scalable growth.

🛵 2️⃣ Prioritize the 2-Wheeler Segment for Rapid Market Capture
📌 **Recommendation:** Focus initial expansion on the 2-wheeler EV segment.
This segment demonstrates higher adoption rates due to affordability and subsidy impact, enabling faster market penetration and strong volume-driven revenue growth.

🔌 3️⃣ Strengthen Charging Infrastructure Through Strategic Partnerships
📌 **Recommendation:** Collaborate with charging infrastructure providers and state authorities in high-demand regions.
Expanding charging networks will reduce range anxiety, increase customer confidence, and accelerate EV adoption.

Thank You

Thank you for your time and attention.
This analysis provided insights into EV adoption trends, government incentives, infrastructure impact, and strategic recommendations for market expansion.
I truly appreciate the opportunity to present these findings and look forward to your feedback and discussion.

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